📐 Pre-Calculus Study Guide

StudyLab — Comprehensive Review

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Unit 1: Functions & Their Properties

What Is a Function?

A `function` is a relation where each input (x) maps to exactly one output (y). We write **f(x)** to denote the output.

```
f(x) = 2x + 3 → input 4 → f(4) = 2(4) + 3 = 11
```

Domain & Range

The `domain` is the set of all valid inputs. The `range` is all possible outputs.

Common domain restrictions:

- **Fractions:** Denominator ≠ 0
- **Square roots:** Expression under radical ≥ 0
- **Logarithms:** Argument > 0

> 💡 To find the domain, ask: "What x-values would break this function?" Then exclude those values.

Types of Functions

Type	Form	Example
Linear	$f(x) = mx + b$	$f(x) = 3x - 1$
Quadratic	$f(x) = ax^2 + bx + c$	$f(x) = x^2 - 4x + 3$
Absolute Value	$f(x) = \|ax + b\|$	$f(x) = \|x - 2\|$
Piecewise	Different rules per interval	$f(x) = \{ x^2 \text{ if } x < 0;\ 2x \text{ if } x \geq 0 \}$

Transformations

- **$f(x) + k$** → shift UP by k
- **$f(x) - k$** → shift DOWN by k
- **$f(x - h)$** → shift RIGHT by h
- **$f(x + h)$** → shift LEFT by h
- **$-f(x)$** → reflect over x-axis
- **$f(-x)$** → reflect over y-axis
- **$a \cdot f(x)$** where $|a| > 1$ → vertical stretch; $0 < |a| < 1$ → compression

> ⚠️ Horizontal shifts are counterintuitive: $f(x - 3)$ shifts RIGHT, not left.

Even & Odd Functions

- **Even:** $f(-x) = f(x)$ — symmetric about y-axis (e.g., x^2, $\cos x$)
- **Odd:** $f(-x) = -f(x)$ — symmetric about origin (e.g., x^3, $\sin x$)

Unit 2: Polynomial & Rational Functions

Polynomials

```
f(x) = aₙxⁿ + aₙ₋₁xⁿ⁻¹ + … + a₁x + a₀
```

The **degree** is the highest exponent. The **leading coefficient** is a_n.

End Behavior

Degree	Leading Coeff	Left End	Right End
Even	Positive	↑ up	↑ up
Even	Negative	↓ down	↓ down
Odd	Positive	↓ down	↑ up
Odd	Negative	↑ up	↓ down

Zeros & Multiplicity

- A polynomial of degree n has at most n real zeros
- Odd multiplicity → graph crosses the x-axis
- Even multiplicity → graph touches and bounces back

Remainder & Factor Theorems

- **Remainder Theorem:** When f(x) ÷ (x – c), remainder = f(c)
- **Factor Theorem:** (x – c) is a factor iff f(c) = 0

Rational Functions: f(x) = P(x)/Q(x)

- **Vertical Asymptotes:** Where Q(x) = 0 and doesn't cancel
- **Horizontal Asymptote:** Compare degrees of P and Q
- **Holes:** Where a factor cancels in both numerator and denominator

```
f(x) = (x−2)(x+1) / (x−2)(x−3) → Hole at x=2, VA at x=3
```

Unit 3: Exponential & Logarithmic Functions

Exponential Functions

```
f(x) = a · bˣ (b > 0, b ≠ 1)
```

- $b > 1$ → exponential growth
- $0 < b < 1$ → exponential decay
- y-intercept always = a (since $b^0 = 1$)
- Horizontal asymptote: $y = 0$

The Number e

e ≈ 2.71828 — base of the natural exponential function.

```
Continuous compound interest: A = Pe^(rt)
```

Logarithms

```
y = log_b(x) means b^y = x
```

log(x) = $\log_{10}(x)$ (common) | **ln(x)** = $\log_e(x)$ (natural)

Properties of Logarithms

Property	Formula
Product Rule	log_b(MN) = log_b(M) + log_b(N)
Quotient Rule	log_b(M/N) = log_b(M) − log_b(N)
Power Rule	log_b(M^p) = p · log_b(M)
Change of Base	log_b(x) = ln(x) / ln(b)

💡 "Logs turn multiplication into addition, division into subtraction, and exponents into multiplication."

Solving Equations

```
Exponential: 3ˣ = 81 → x = ln(81)/ln(3) = 4
```

```
Logarithmic: log₂(x) = 5 → x = 2⁵ = 32
```

⚠️ Always check solutions in log equations — you can't take the log of a negative number or zero.

Unit 4: Trigonometry Fundamentals

Radians & Degrees

π radians = 180° | radians = degrees × (π/180) | degrees = radians × (180/π)

Degrees	0°	30°	45°	60°	90°	180°	360°
Radians	0	π/6	π/4	π/3	π/2	π	2π

Unit Circle Key Values

θ	sin θ	cos θ	tan θ
0°	0	1	0
30°	1/2	√3/2	√3/3
45°	√2/2	√2/2	1
60°	√3/2	1/2	√3
90°	1	0	undef

💡 Mnemonic for sin: √0/2, √1/2, √2/2, √3/2, √4/2 → simplifies to 0, 1/2, √2/2, √3/2, 1

SOH-CAH-TOA

sin = Opposite/Hypotenuse | cos = Adjacent/Hypotenuse | tan = Opposite/Adjacent

Signs by Quadrant — "All Students Take Calculus"

- **QI:** All positive | **QII:** Sin+ | **QIII:** Tan+ | **QIV:** Cos+

Graphing: y = A sin(Bx – C) + D

- Amplitude = |A| | Period = 2π/|B| | Phase shift = C/B | Vertical shift = D

Unit 5: Trig Identities & Equations

Pythagorean Identities

```
sin²θ + cos²θ = 1 | 1 + tan²θ = sec²θ | 1 + cot²θ = csc²θ
```

Sum & Difference Formulas

```
sin(A ± B) = sin A cos B ± cos A sin B
```

```
cos(A ± B) = cos A cos B ∓ sin A sin B
```

```
tan(A ± B) = (tan A ± tan B) / (1 ∓ tan A tan B)
```

⚠️ Note the sign flip in cosine: cos(A – B) = cos A cos B + sin A sin B

Double-Angle Formulas

```
sin(2θ) = 2 sin θ cos θ
```

```
cos(2θ) = cos²θ − sin²θ = 2cos²θ − 1 = 1 − 2sin²θ
```

```
tan(2θ) = 2tan θ / (1 − tan²θ)
```

Verifying Identities — Strategy

1. Work with ONE side (usually the more complex side)
2. Convert everything to sin and cos
3. Factor when possible
4. Use Pythagorean identities to substitute

Solving Trig Equations

1. Isolate the trig function
2. Find the reference angle
3. Find all solutions in [0, 2π)
4. Add general solution: + 2nπ (sin/cos) or + nπ (tan)

Unit 6: Sequences, Series & Limits

Arithmetic Sequences

```
aₙ = a₁ + (n − 1)d | Sₙ = n/2 · (a₁ + aₙ)
```

Example: 3, 7, 11, 15 → d = 4, a_{10} = 3 + 9(4) = 39

Geometric Sequences

```
aₙ = a₁ · r^(n−1) | Sₙ = a₁(1 − rⁿ)/(1 − r)
```

Infinite series (converges when |r| < 1):

```
S∞ = a₁ / (1 − r)
```

💡 An infinite geometric series only has a finite sum when |r| < 1.

Sigma Notation

```
∑(i=1 to n) i = n(n+1)/2 | ∑(i=1 to n) i² = n(n+1)(2n+1)/6
```

Introduction to Limits

```
lim(x→c) f(x) = L means "as x approaches c, f(x) approaches L"
```

Key techniques:

- **Direct substitution:** If f(c) is defined, lim = f(c)
- **Factoring:** If you get 0/0, factor and cancel
- **Limits at infinity:** Divide by highest power of x in denominator

```
lim(x→2) (x²−4)/(x−2) = lim(x→2) (x+2) = 4
```

⚠️ A limit can exist even if the function is undefined at that point.